EEE3096S Practical 1A

STM32 GPIO, Timers, and Interrupts

Student Name 1, Student Number 1
Student Name 2, Student Number 2

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Notes on This Template

This template covers every task in Practical 1A. Delete this section before you submit.

Do not paste your code into this report. Your tutor views your code on GitHub. Each task carries marks for your written answers and marks for your demonstration. During your demo, your tutor asks a demo question on the spot. Know your implementation well, since you answer it without notes.

Name your final PDF exactly as the practical instructs: prac_01A_stdnum001_stdnum002.pdf, with both student numbers in lowercase.

1 Hardware Orientation

1. GPIO pins connected to the 8 user LEDs: [Your answer]
2. GPIO pins connected to the 4 push buttons: [Your answer]
3. Pins connected to the 8 MHz crystal: [Your answer]
4. Default system clock frequency using the internal HSI oscillator: [Your answer]
5. Purpose of the loopback header P2: [Your answer]

2 Task 1: Timer Configuration and Verification

2.1 Predictions

Record these before you write any code or calculate any values.

- System clock frequency: [Your answer]
- Expected prescaler value: [Your answer]
- Expected ARR value: [Your answer]
- Expected interrupt period (s): [Your answer]

2.2 Calculations

Show your working for the timer clock frequency and the ARR value.

$$\text{Timer Clock} = \frac{\text{System Clock}}{\text{Prescaler} + 1} = \text{[Your working]}$$

$$\text{ARR} = \text{[Your working]}$$

2.3 Measurement and Verification

- Measured period from the oscilloscope: [Your answer]
- Comparison with your expected period: [Your answer]

3 Task 2: LED Control with Timer Interrupt

Confirm your running light video is in your submission folder, or link it here: [Your answer]

4 Task 3: Button Input and Software Debounce

4.1 Records

- Expected switch bounce duration: [Your answer]
- Maximum bounce duration measured on the oscilloscope: [Your answer]
- Chosen software debounce delay time: [Your answer]
- Formula to calculate the ARR value from the desired period in milliseconds: [Your answer]

4.2 Question 4

When you update the ARR register dynamically, does the timer frequency change immediately mid-cycle, or does the timer wait for the next update event? Base your answer on the STM32 timer preload register behaviour.

[Your answer]

5 Task 4: Multi-Mode LED Control

5.1 State Machine Explanation

Explain your state machine implementation for Mode 3. Describe the SPARKLE_IDLE, SPARKLE_DISPLAY and SPARKLE_TURN_OFF states, and how your code moves between them.

[Your answer]

5.2 Measurements

- Measured minimum LED-on time for Mode 3: [Your answer]
- Measured maximum LED-on time for Mode 3: [Your answer]

6 Task 5: ADC Interrupts and LED Level Meter

6.1 Records

- Predicted 12-bit ADC register value, in decimal, for a 2.0 V input: [Your answer]
- Measured multimeter voltage at the moment the 5th LED turns on: [Your answer]
- Calculation comparing your measured voltage against the theoretical threshold: [Your answer]

6.2 Question

If a student accidentally sets the signal generator to output a 5.0 V DC signal into PA5, what physical damage might occur? Consult the I/O structure column for PA5 in the STM32F051x datasheet.

[Your answer]

7 Task 6: Software PWM via Timer Interrupts

7.1 Records

- Calculated Timer Prescaler and ARR values for exactly 10 kHz: [Your answer]
- Measured total period of the signal: [Your answer]
- Measured positive pulse width: [Your answer]

7.2 Question

A project requires a 50 Hz signal with a duty cycle resolution of 1 percent. What is the exact timer interrupt frequency required? Show your calculation.

[Your answer]